

Development of Finger Braille Recognition System*

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Abstract

Finger Braille is one of the communication media of deafblind people. In one-handed Finger Braille, a sender dots the left part of the Braille code on the distal interphalangeal (DIP) joints of the index, middle and ring fingers of a receiver, and subsequently dots the right part of the Braille code on the proximal interphalangeal (PIP) joints of the same fingers. Because there is a small number of non-disabled people who are skilled in Finger Braille, deafblind people communicate in this medium only through an interpreter. In this study, we developed a Finger Braille recognition system using small piezoelectric accelerometers worn by the receiver. We first conducted a measurement experiment to derive algorithms for the recognition of the dotted fingers and positions. The results showed that the accuracy of the recognition of dotted fingers was 92.9%, and the recognition system was independent of the dotted strength, dotted position and receiver. The accuracy of the recognition of dotted positions was 81.9%, and the recognition system could recognize dotted positions if the receiver's hand formed a natural longitudinal arch on the desk. Next, an evaluation experiment was carried out. The results showed that the accuracy of the recognition of dotted fingers by dotting was 89.7%, and the accuracy of recognition of dotted positions by dotting was 92.3%. Therefore, the recognition system could recognize sentences accurately when the interpreter dotted clearly.

Key words: Finger Braille, Deafblind, Recognition System, Communication Aid, Accelerometer, Shock Acceleration, Recognition of Dotted Fingers, Recognition of Dotted Positions

1. Introduction

The Ministry of Health, Labour and Welfare of Japan estimates that there are nearly 22,000 deafblind people in Japan (2006). Communication is one of their largest barriers to independent living and participation. Deafblind people use many different communication media, depending on the age of onset of deafness and blindness and the available resources. “Yubi-Tenji” (Finger Braille) is one of the tactual communication media utilized by deafblind individuals (see Fig. 1). In two-handed Finger Braille, the sender's index finger,

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middle finger and ring finger of both hands function like the keys of a Braille typewriter. The sender dots Braille code on the fingers of the receiver. The receiver is assumed to recognize the Braille code. In one-handed Finger Braille, the sender dots the left part of Braille code on the distal interphalangeal (DIP) joints of the three fingers of the receiver, and then the sender dots the right part of Braille code on the proximal interphalangeal (PIP) joints. Deafblind people who are skilled in Finger Braille can communicate words and express various emotions because of the prosody (intonation) of Finger Braille ⁽¹⁾. Because there is such a small number of non-disabled people who are skilled in Finger Braille, deafblind people communicate only through an interpreter. Thus, the participation of deafblind people is greatly restricted.

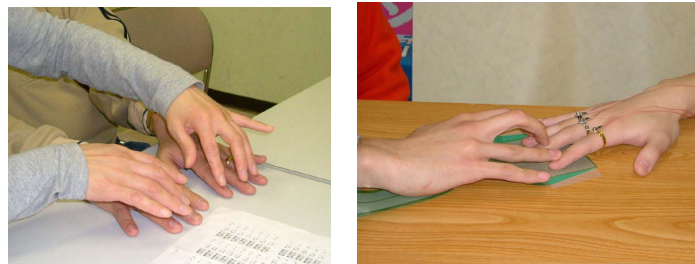


Fig. 1 Two-handed Finger Braille (left) and one-handed Finger Braille (right)

Various Finger Braille input devices, including a wearable input device, have been developed. Uehara et al. developed a Finger Braille glove system with accelerometers mounted on the fingertips ⁽²⁾. Fukumoto et al. developed a wearable input device with accelerometers mounted on the top of rings ⁽³⁾. Hoshino et al. developed a Finger Braille input system that mounted accelerometers on the middle phalanges ⁽⁴⁾. In addition, Ochi et al. developed a bracelet-type Finger Braille input device with eighteen mounted accelerometers ⁽⁵⁾. These devices require deafblind people to wear gloves, rings or bracelets to input Finger Braille. With these support devices, deafblind people are not only burdened with wearing the sensors, but also they must master a new communication system using such support devices.

The objective of this study is the development of a Finger Braille support device which employs communication through skin contact, because skin contact is the only non-verbal communication possible for deafblind people. The concept of the proposed Finger Braille support device is shown in Fig. 2. The advantages of this support device are as follows: both deafblind people and non-disabled people who are not skilled in Finger Braille can communicate using conventional Finger Braille, and all sensors are worn by the non-disabled people. This support device consists of a Finger Braille teaching system and a Finger Braille recognition system. The teaching system recognizes the speech of non-disabled people and displays the associated dot pattern of Finger Braille. Non-disabled people can then dot Finger Braille on the fingers of deafblind people by observing the displayed dot pattern ⁽⁶⁾. The recognition system recognizes the dotting of Finger Braille by the deafblind people and synthesizes the speech for non-disabled people. Thus, deaf-blind people can communicate without being encumbered by the support device.

To develop the recognition system, some requirements must be fulfilled.

In Finger Braille, the sender dots Braille codes directly on the fingers of the receiver. A rule of Finger Braille is that the sender constantly touches the fingers of the receiver even when not dotting, because the receivers feel uncomfortable in the absence of touching or tactile cues. Therefore, sensors must not hinder the skin contact between deafblind people and non-disabled people. Because of the lack of visual and audio information, deafblind people experience difficulty in mastering a new communication system. Thus, the sensors

must be worn by the receiver (non-disabled people). In this study, we adopted small accelerometers mounted on the top of finger rings.

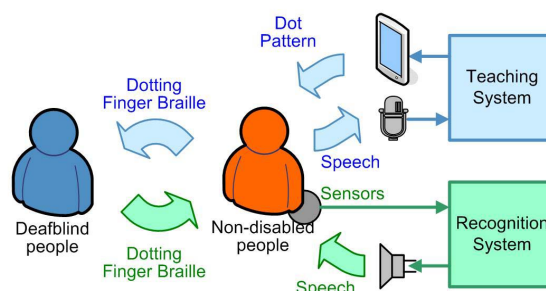


Fig. 2 Concept of the proposed Finger Braille support device

In our concept of assistance, deafblind people are equipped with the recognition system, which non-disabled people can also use. The non-disabled people are unspecified. Thus, the recognition system must be independent of the receiver.

For prosody of Finger Braille, the sender dots long and strong at the end of clauses and sentences. The sender can dot strongly with anger, or weakly with sadness ⁽¹⁾. Thus, the recognition system must be independent of dotted strength.

To develop the recognition system, we adopted one-handed Finger Braille. Here, the recognition system requires independence of the dotted position and recognition of the dotted positions.

In this paper we describe the Finger Braille recognition system and present experimental results. We first conducted a measurement experiment to derive algorithms for the recognition of dotted fingers and positions. Then, an evaluation experiment was carried out.

2. Configuration of Finger Braille recognition system

Figure 3 shows the configuration of the Finger Braille recognition system. The non-disabled people wore rings with small piezoelectric accelerometers (yamco 10SW, Yamaichi Electronics) on the index finger, middle finger and ring finger. Each accelerometer was mounted on the top of the ring. The accelerometers were connected to a tablet PC (TC1100, HP) through charge amplifiers (yamco 4101, Yamaichi Electronics) and an A/D converter (USB-9215A-BNC, National Instruments). The sampling frequency was 10 kHz, the measurement range was $\pm 250 \text{ m/s}^2$, and the sensibility was 0.2 m/s^2 .

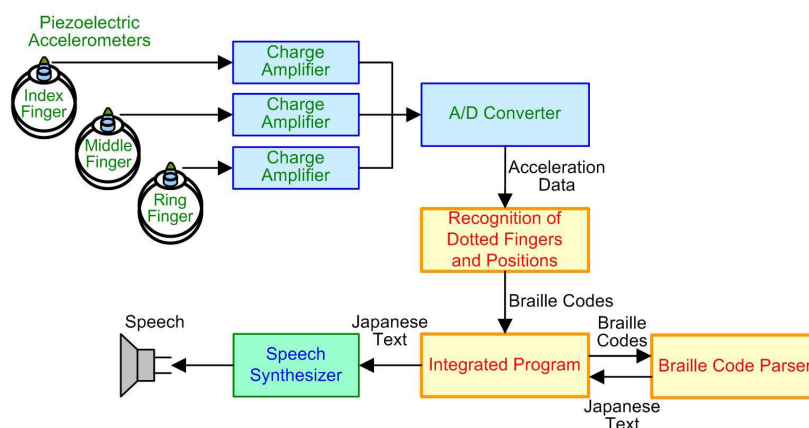


Fig. 3 Configuration of the recognition system

First, the accelerometers detected the accelerations of the dotting, and acceleration data were acquired. Second, the recognition system recognized the dotted fingers and positions. Third, by parsing the recognized Braille codes, the recognition system converted the Braille codes to Japanese text. Finally, the recognition system synthesized the speech of the Japanese text.

The operating system (OS) was Microsoft Windows XP. The programs of recognition of the dotted fingers and positions were programmed in LabVIEW 8.0 (National Instruments). The Braille code parser was programmed in Win-Prolog 4.500 (Logic Programming Associates). The integrated program was programmed in Microsoft Visual Basic 6. The speech synthesizer was VoiceText (Pentax).

3. Recognition of the dotted fingers

3.1 Methods of the measurement experiment

To derive the algorithms for the recognition of the dotted fingers and positions, a measurement experiment of Finger Braille dotting was conducted ⁽⁷⁾.

The subjects (receivers) were ten non-disabled college students (five males and five females). All subjects gave informed consent after hearing the description of the study. A tester (sender) was not a Finger Braille interpreter but was skilled in this technique.

The procedure was as follows. The subject wore the accelerometers. The tester dotted seven characters ($\begin{smallmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{smallmatrix}$) on the fingers of the subject with three dotting conditions (normal strength, weak strength and strong strength). The subject's hand was set on the desk in each dotting condition. Five experimental sessions of each dotting condition were conducted.

3.2 Features of shock acceleration by dotting

Figure 4 shows the features of the accelerations by dotting. When the sender dotted Finger Braille on the fingers of the receiver, the accelerometers detected shock accelerations by the dotting of the mounted finger (self dotting) and shock accelerations by the dotting of the other fingers (cross talk). Figure 5 shows the shock accelerations by self dotting and cross talk. The accelerations by cross talk were less than the accelerations by self dotting. *AI*, *AM* and *AR* indicate the amplitudes of accelerations of the index, middle and ring fingers, respectively. The results of the measurement experiment showed that the mean of the amplitude of the acceleration by self dotting was 71.3 m/s^2 . The mean of the amplitude of acceleration by cross talk was 11.0 m/s^2 . The maximum amplitude of acceleration by cross talk was 74.6 m/s^2 . The minimum amplitude of acceleration by self dotting was 1.6 m/s^2 . The range of the amplitude of acceleration by self dotting and the range of the amplitude of acceleration by cross talk overlap each other. Thus, it is difficult to recognize the acceleration by self dotting by using a constant threshold for the amplitude ⁽²⁾⁽⁷⁾.

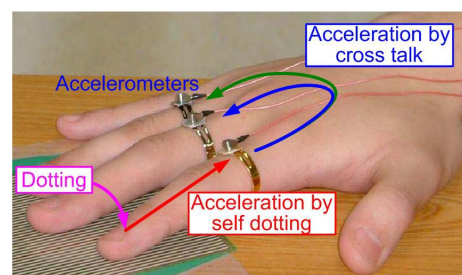


Fig. 4 Features of the accelerations by dotting

Figure 6 shows the frequency spectrums of the accelerations by self dotting and cross

talk. The acceleration data for 100 ms (pre-trigger 20 ms and post-trigger 80 ms) were recorded. The window function was the Hanning window. The difference of power between self dotting and cross talk was greater at approximately 100 Hz ⁽³⁾⁽⁴⁾. *PI*, *PM* and *PR* indicate the powers at 100 Hz of the index, middle and ring fingers, respectively. The mean of the power at 100 Hz by self dotting was -36.5 dB Vrms. The mean of the power at 100 Hz by cross talk was -60.0 dB Vrms. The maximum power at 100 Hz by cross talk was -32.2 dB Vrms, and the minimum power at 100 Hz by self dotting was -76.2 dB Vrms. The range of the power at 100 Hz by self dotting and the range of the power at 100 Hz by cross talk overlap each other. Thus, it is also difficult to recognize the acceleration by self dotting using a constant threshold of power at 100 Hz ⁽⁷⁾.

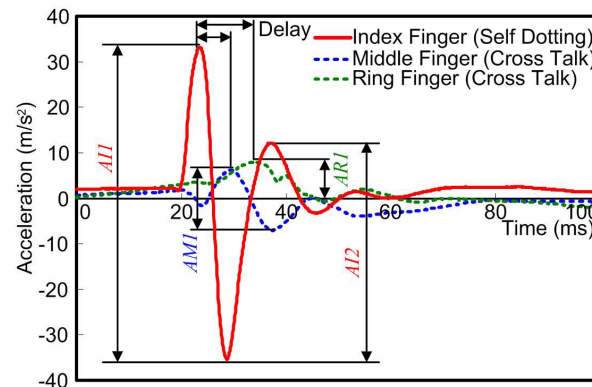


Fig. 5 Shock accelerations by self dotting and cross talk (dotting on DIP joints: hard impact)

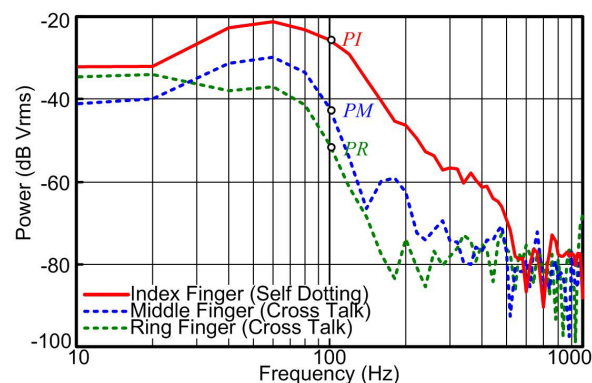


Fig. 6 Frequency spectrums of accelerations by self dotting and cross talk

Because the accelerations by cross talk must have a delay (adjacent fingers: 5.0 ms, index finger and ring finger: 8.9 ms), the first detected acceleration must be the acceleration by self dotting. In the case of Fig. 5, the acceleration of the index finger is the first detected acceleration. Then by setting the acceleration of the index finger as the dynamic threshold, the recognition system can recognize the acceleration of the middle finger and ring finger. We noted two parameters related to the index finger, the amplitude of acceleration (*AII*) and the power at 100 Hz (*PI*).

3.3 Algorithm for the recognition of the dotted fingers

We derived an algorithm for the recognition of dotted fingers. For example, the variables and equations in the case of Figs. 5 and 6 are identified.

Step 1: Acquire the acceleration data for 100 ms (pre-trigger 20 ms and post-trigger 80 ms) when the shock accelerations by dotting are detected.

Step 2: Set the amplitude and power at 100 Hz of the first detected acceleration as the dynamic thresholds (index finger: $AI1$, PI).

Step 3: If the amplitude of the second detected acceleration is greater than half of the amplitude of the first detected acceleration (middle finger: $AMI > AI1/2$) or the power at 100 Hz of the second detected acceleration is greater than the power at 100 Hz of the first detected acceleration minus 10 dB Vrms (middle finger: $PM > PI-10$), the second detected acceleration is recognized as the acceleration by self dotting.

Step 4: If the amplitude of the second detected acceleration is less than or equal to half of the amplitude of the first detected acceleration (middle finger: $AMI \leq AI1/2$) and the power at 100 Hz of the second detected acceleration is less than or equal to the power at 100 Hz of the first detected acceleration minus 10 dB Vrms (middle finger: $PM \leq PI-10$), the second detected acceleration is recognized as the acceleration by cross talk.

Step 5: If the power at 100 Hz of the second detected acceleration is less than -58 dB Vrms (middle finger: $PM < -58$), the second detected acceleration is recognized as the acceleration by cross talk.

Step 6: Steps 3~5 apply to the third detected acceleration (ring finger: ARI , PR).

3.4 Accuracy of the recognition of the dotted fingers

Applying the algorithm of the recognition of dotted fingers, all acceleration data by self dotting or cross talk were estimated. The overall accuracy of the recognition of dotted fingers was 92.9%.

Because the accuracies of all dotting conditions and dotted positions were greater than 90%, the recognition system was independent of the dotted strength and dotted position.

For one male subject (subject 3), the power at 100 Hz of the middle finger was less 5 dB Vrms in the sessions of normal strength and weak strength. When the acceleration of the middle finger was the first detected acceleration, the accelerations by cross talk of the index finger were misrecognized. Consequently, the accuracies of the middle finger and middle + ring fingers were less than the other accuracies. This phenomenon occurred only in the sessions of normal and weak strengths. After these sessions, in the sessions of strong strength, the power at 100 Hz of the middle finger improved. Every subject re-set their hand on the desk in each condition. Thus, it was considered that this phenomenon occurred by the hand posture, and re-setting the hand resulted in improved power.

For one female subject (subject 7), the accuracy was less than the accuracies of the other subjects. The experiment was carried out for male subjects first and then for female subjects. Because the hand of subject 7 was smaller than the hands of the male subjects, the tester could not dot Finger Braille stably. Thus, the accuracy of subject 7 was lower. Considering the results of all the subjects, the recognition system was independent of the receiver.

As mentioned above, Uehara et al. developed the Finger Braille glove system, which had mounted accelerometers on the fingertips⁽²⁾. Uehara et al. used a constant threshold of amplitude of acceleration for the recognition of dotting. Fukumoto et al. developed the wearable input device that mounted accelerometers on the top of rings⁽³⁾. Fukumoto et al. used a constant threshold of power at approximately 100 Hz for the recognition of dotting. Hoshino et al. developed the Finger Braille input system that mounted accelerometers on the middle phalanges⁽⁴⁾. In addition, Hoshino et al. used a constant threshold of power at 40~100 Hz for the recognition of dotting.

To compare this study with these previous studies, constant thresholds of the amplitude of acceleration and power at 100 Hz were calculated by applying decision methods for the constant thresholds (for Uehara's method and Fukumoto's method). Using these thresholds, all acceleration data by self dotting or cross talk in our study were recalculated. The estimated accuracy using the constant thresholds of the amplitude of acceleration was

76.1%. The estimated accuracy using the constant thresholds of power at 100 Hz was 79.0%. However, the accelerometers were worn by the senders in the previous studies and by the receivers in this study, and so our proposed algorithm of the recognition of dotted fingers is suitable for this study's recognition system.

4. Recognition of the dotted positions

4.1 Features of shock acceleration by dotted positions

We analyzed the features of shock acceleration from the results of the measurement experiment (§3). Figure 7 shows the features of the accelerations by the dotted positions. When the receiver's hand forms a natural longitudinal arch on the desk, the DIP joints are close to the desk, and space exists under the PIP joints. Dotting on the DIP joints causes a hard impact, and dotting on the PIP joints causes a soft impact. Figure 5 shows the shock accelerations by dotting on the DIP joints. Figure 8 shows the shock accelerations by dotting on the PIP joints. $AI1$ indicates the amplitude of acceleration of the index finger. $AI2$ indicates the damping amplitude of the index finger. The difference of each impact is indicated by its damping amplitude. The damping amplitude ratio of accelerations by dotting on the DIP joints ($AI2/AI1$) is greater than the damping amplitude ratio of accelerations by dotting on the PIP joints.

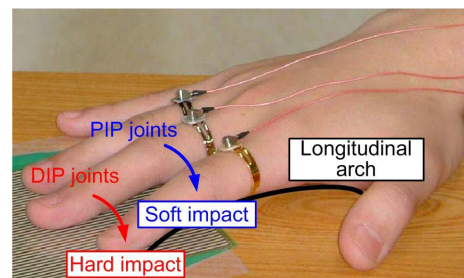


Fig. 7 Features of the accelerations by dotted positions

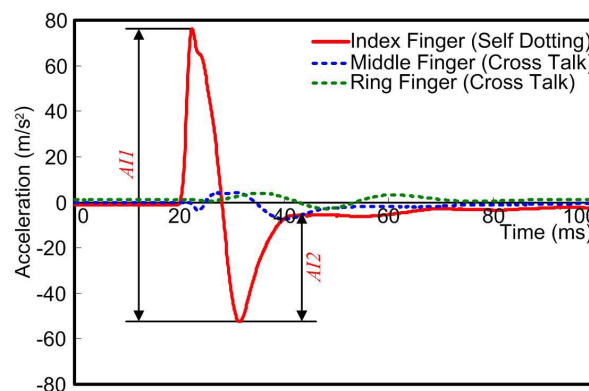


Fig. 8 Shock accelerations by self dotting and cross talk (dotting on PIP joints: soft impact)

4.2 Algorithm for the recognition of the dotted positions

We derived the following algorithm for the recognition of the dotted positions. For this example, the variables and equations for the cases of Figs. 5 and 8 were calculated.

Step 1: Calculate the damping amplitude ratio ($AI2/AI1$) of the acceleration by self dotting.

Step 2: If the damping amplitude ratio is greater than 0.5 ($AI2/AI1 > 0.5$), the

acceleration is recognized as the accelerations by dotting on the DIP joints.

Step 3: If the damping amplitude ratio is less than or equal to 0.5 ($AI2/AI1 \leq 0.5$), the acceleration is recognized as the accelerations by dotting on the PIP joints.

Step 4: If the amplitude of the acceleration is greater than 150 m/s^2 ($AI1 > 150$), the acceleration is recognized as the accelerations by dotting on the PIP joints.

Step 5: If two or three fingers are dotted at the same time (self dotting), the mean of the damping amplitude ratios is calculated and Steps 2~4 are applied.

4.3 Accuracy of the recognition of the dotted positions

Applying the algorithm of the recognition of the dotted positions, the accelerations by dotting on the DIP joints or PIP joints were estimated by using the acceleration data of §3. The overall accuracy of the recognition of the dotted positions was 81.9%.

The accuracies of the PIP joints of one male subject (subject 3) and two female subjects (subjects 6 and 7) were less than the other accuracies. For subject 3, it was considered that the accuracy was less due to the abovementioned hand posture (§3). For subjects 6 and 7, their hands were flexible and fully contacted the desk. Because their hands did not form the natural longitudinal arch on the desk, dotting on the PIP joints caused a hard impact. Thus, the accuracies of the PIP joints of subjects 6 and 7 were lower than the other accuracies. Therefore, it was considered that the recognition system could recognize dotted positions if the receiver's hand formed the natural longitudinal arch on the desk.

5. Braille code parser and integrated program

We developed a Braille code parser by applying the technology of natural language processing (BUP system) ⁽⁸⁾. The Braille code parser parsed the Braille code and converted it into Japanese text. The Braille code parser consisted of a dictionary, grammar, BUP translator and control program. The dictionary and the grammar were described in definite clause grammar (DCG). The BUP translator translated the dictionary and grammar into a Prolog program. The control program controlled the execution of parsing.

When the programs for the recognition of dotted fingers and positions recognize a dotting, the integrated program sends a list of Braille code to the Braille code parser. Then the Braille code parser parses the list of Braille code. If the list of Braille code is grammatically correct, the Braille code parser sends the converted Japanese text to the integrated program. If the list of Braille code is grammatically incorrect, the Braille code parser sends a "no" to the integrated program.

Finally, when the integrated program receives the Japanese text from the Braille code parser, the integrated program allows the speech synthesizer to synthesize the Japanese text.

6. Evaluation experiment

6.1 Methods

To evaluate the recognition of sentences dotted by the Finger Braille interpreter, an evaluation experiment of sentence recognition was carried out.

The subject (sender) was a non-disabled Finger Braille interpreter (experiment: 22 years). The subject gave informed consent after hearing a description of the study.

The dialogues consisted of four daily conversations in a Japanese textbook for foreign beginners ⁽⁹⁾ (total: 51 sentences, 143 clauses, 288 words, 686 characters). The numbers of the dottings of the dialogues are listed in Table 1. In Finger Braille, some characters are dotted on both the DIP joints and PIP joints and some characters are dotted only on the DIP joints or the PIP joints. The average of dotted times per character was 1.75.

The procedure was as follows. A tester wore the accelerometers. The subject dotted the

sentences in each dialogue clearly. The tester's hand set on the desk in each conversation and formed the natural longitudinal arch. If the recognition system synthesized the misrecognized speech, the subject would stop dotting or re-dot the dialogues. To prevent unnecessary pause or re-dotting by the subject, the speech synthesizer was turned off during the experiment. The lists of the recognized Braille code were recoded in the hard disk drive of the recognition system.

Table 1 Numbers of the dottings of the dialogues

Dotted fingers	Dotting on DIP joints	Dotting on PIP joints
Index finger	152	71
Middle finger	77	161
Ring finger	70	66
Index + middle fingers	110	84
Middle + ring fingers	54	126
Index + ring fingers	72	39
Index + middle + ring fingers	70	72

6.2 Results

The mean of the dotting speed was 37.0 characters/min. This was almost 1/3 of the normal dotting speed.

To evaluate the accuracy of recognition, we checked the lists of the recognized Braille code and calculated the accuracies of the recognition by dotting (each dotting on DIP joints or PIP joints) and by character (one or two dottings). Figure 9 shows the accuracies of the recognition of dotted fingers as a function of conversation and as a function of the calculation unit. Figure 10 shows the accuracies of the recognition of dotted fingers by dotting as a function of the dotted fingers and as a function of the dotted positions.

The overall accuracy of the recognition of dotted fingers by dotting was 89.7%. In the experiment of conversation 3, the power at 100 Hz of the middle finger was less 5 dB Vrms. The accuracy of conversation 3 was 77.2%. The accuracy without conversation 3 was 94.3%. The accuracies of the middle finger and middle + ring fingers of the dotting on the PIP joints were less than the other accuracies; the accuracies of the index + middle fingers and middle + ring fingers of the dotting on the PIP joints were also less than the other accuracies.

The overall accuracy of the recognition of dotted fingers by character was 82.6%. The accuracy without conversation 3 was 90.0%.

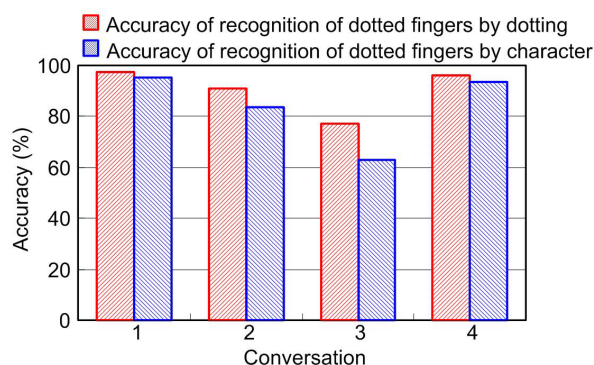


Fig. 9 Accuracies of the recognition of dotted fingers as a function of conversation and as a function of the calculation unit

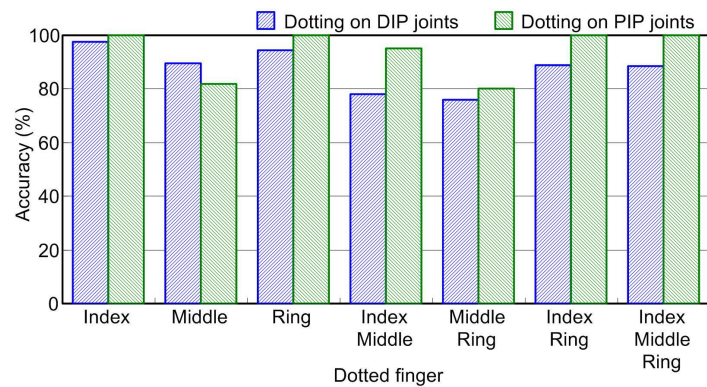


Fig. 10 Accuracies of the recognition of dotted fingers by dotting as a function of the dotted fingers and as a function of the dotted positions

Figure 11 shows the accuracies of the recognition of dotted positions as a function of conversation and as a function of the calculation unit. Figure 12 shows the accuracies of the recognition of dotted positions by dotting as a function of the dotted fingers and as a function of the dotted positions.

The overall accuracy of the recognition of dotted positions by dotting was 92.3%. The accuracy of conversation 3 was 88.8%. The accuracy without conversation 3 was 94.9%. The accuracies of the dotting on the PIP joints of the index finger and middle finger were less than the other accuracies.

The overall accuracy of the recognition of dotted positions by character was 88.3%. The accuracy without conversation 3 was 91.2%.

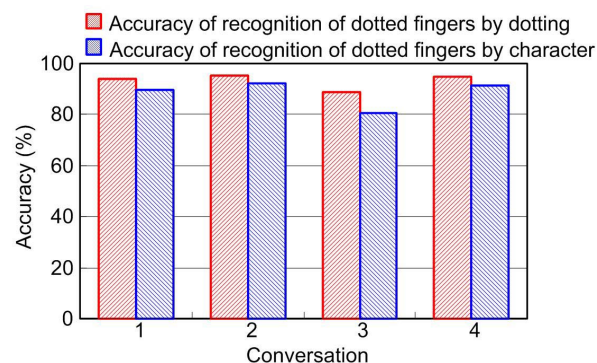


Fig. 11 Accuracies of the recognition of dotted positions as a function of conversation and as a function of the calculation unit

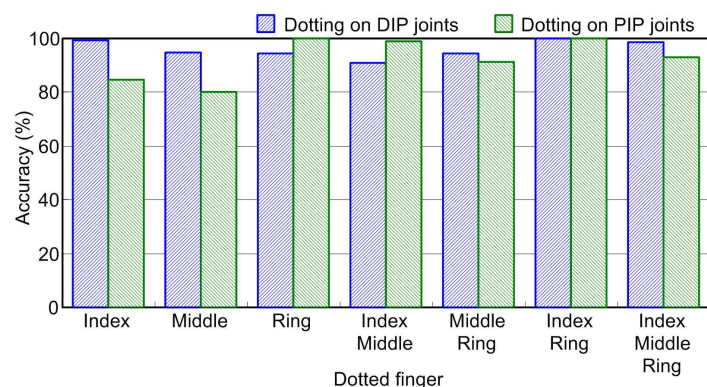


Fig. 12 Accuracies of the recognition of dotted positions by dotting as a function of the dotted fingers and as a function of the dotted positions

6.3 Discussion

The accuracy of the recognition of dotted fingers by dotting without conversation 3 was 94.3%, and the accuracy of the recognition of dotted positions by dotting without conversation 3 was 94.9%. The accuracy of the recognition of dotted fingers by character without conversation 3 was 90.0%, and the accuracy of the recognition of dotted positions by character without conversation 3 was 91.2%.

In the experiment of conversation 3, the power at 100 Hz of the middle finger was less than 5 dB Vrms, although the power improved in the experiment of conversation 4. This phenomenon was the same as the phenomenon that occurred in the measurement experiment (§3). As real communication using the recognition system, non-disabled people (receiver) can re-set their hand on the desk when they notice a decreased accuracy of recognition. The re-setting of the receiver's hand should be allowed in the communication.

As previously mentioned, Uehara et al. developed a Finger Braille glove system with accelerometers mounted on the fingertips ⁽²⁾. Three Finger Braille interpreters wore the glove system and dotted Finger Braille. The accuracy of recognition was 73.0%. The dialogues that they used were the number of characters; the dotting speed and range of amplitude of acceleration were not clear. But the accuracy of recognition by our recognition system was greater than or equal to the accuracy of recognition by the glove system.

Hoshino et al. developed the Finger Braille input system that mounted accelerometers on the middle phalanges ⁽⁴⁾. Three visually impaired people and two non-disabled people who were skilled in Finger Braille wore the input system and dotted 100 randomized characters. They reported that the accuracy of recognition was 99.3%. Because the characters did not form sentences, the subjects might not express the prosody of Finger Braille.

To compare our study with these previous studies, our recognition system could recognize the sentences accurately when the interpreter dotted clearly. Although the accuracy of the recognition is high, the Braille code parser can not convert the list of the Braille code which was grammatically incorrect into the Japanese text. Then the recognition system can not synthesize the misrecognized clauses. We have been improving the Braille code parser.

In this experiment, the subject pointed out the importance of feedback from the non-disabled person (receiver) to the deafblind person (sender) as to whether the non-disabled person recognized the Finger Braille correctly. Because of their deafness and blindness, deafblind people cannot confirm the synthesized speech and displayed text of the recognition system. Therefore, deafblind people are uneasy without feedback, and they will find it difficult to keep up with the conversation. Therefore, we have been developing the combination of a recognition system and teaching system to display the dot pattern of the recognized sentence, so that receivers can offer feedback to senders.

7. Conclusion

In this paper, we developed a Finger Braille recognition system. We conducted a measurement experiment to derive the algorithms for the recognition of dotted fingers and positions. The results of the measurement experiment showed that the accuracy of the recognition of dotted fingers was 92.9%, and the recognition system was independent of the dotted strength, dotted position and receiver. The accuracy of the recognition of dotted positions was 81.9%, and the recognition system could recognize dotted positions if the receiver's hand formed a natural longitudinal arch on the desk. Next, an evaluation experiment was carried out. The results of the evaluation experiment showed that the accuracy of the recognition of dotted fingers by dotting was 89.7% (94.3% without conversation 3), and the accuracy of the recognition of dotted positions by dotting was

92.3% (94.9% without conversation 3). Therefore, the recognition system could recognize sentences accurately when the interpreter dotted clearly. We confirmed that non-disabled people (receiver) should re-set their hand on the desk when they notice a decrease of the accuracy of recognition.

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